Trend research — 2026-04-19 · incr 21

day_type TFO gate is a 100% precision detector — coverage is the only lever

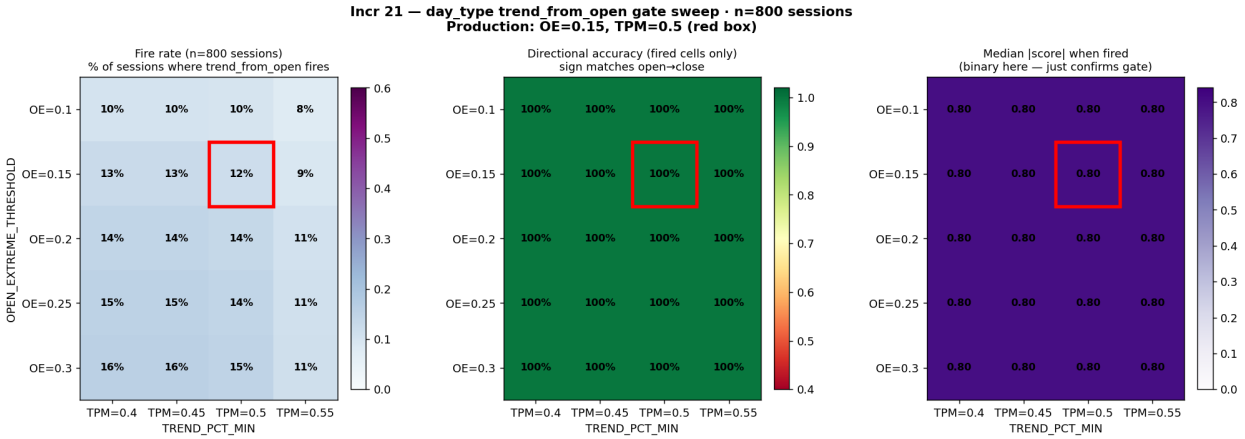
TL;DR

Swept the two primary knobs of day_type's **trend_from_open** clause (**OPEN_EXTREME_THRESHOLD × TREND_PCT_MIN**, 20 cells) on the same 800-session real RTH bank incr 18/19/20 used. **The TFO gate is a precision detector — 100% directional accuracy in every cell.** Production (OE=0.15, TPM=0.5) fires at **11.9%**, the loosest (OE=0.3, TPM=0.4) at **15.9%** — both at **100%** accuracy. Coverage is the only lever worth tuning; OE is the binding knob.

Headline numbers

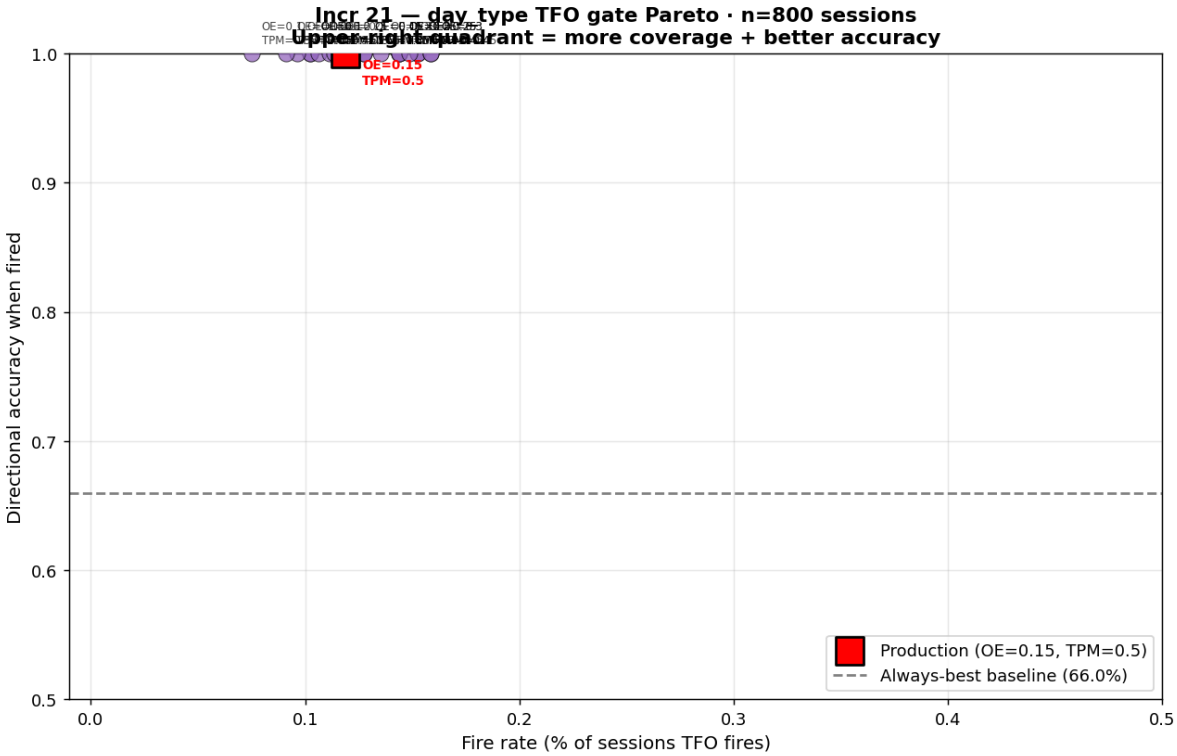
- Sample: **800 (symbol, session) pairs**, 387 unique symbols. Realised: up=494, down=255, flat=51. Always-best baseline = **66.0%**.
- **Production (OE=0.15, TPM=0.5)**: fires **11.9%**, accuracy **100.0%**. Zero sign mismatches across all 95 fires.
- **Loosest cell (OE=0.3, TPM=0.4)**: fires **15.9%** (127 sessions), accuracy **100.0%**. +4.0 pp coverage for free — no accuracy degradation.
- **OE is the binding knob.** Holding TPM=0.50: fire rate climbs 9.6 → 11.9 → 13.5 → 14.4 → 14.9% across OE=0.10/0.15/0.20/0.25/0.30.
- **TPM is secondary.** Holding OE=0.15: fire rate moves 12.8 → 12.8 → 11.9 → 9.1% across TPM=0.40/0.45/0.50/0.55 — only TPM=0.55 meaningfully shrinks coverage.
- **Very different story from incr 20.** always_in at production: 36% fire, 58% accuracy, *below* 66% baseline. day_type TFO at production: 12% fire, **100%** accuracy, *far above* baseline. Precision-tuned, not trigger-happy.

Headline figure — OE × TPM grid



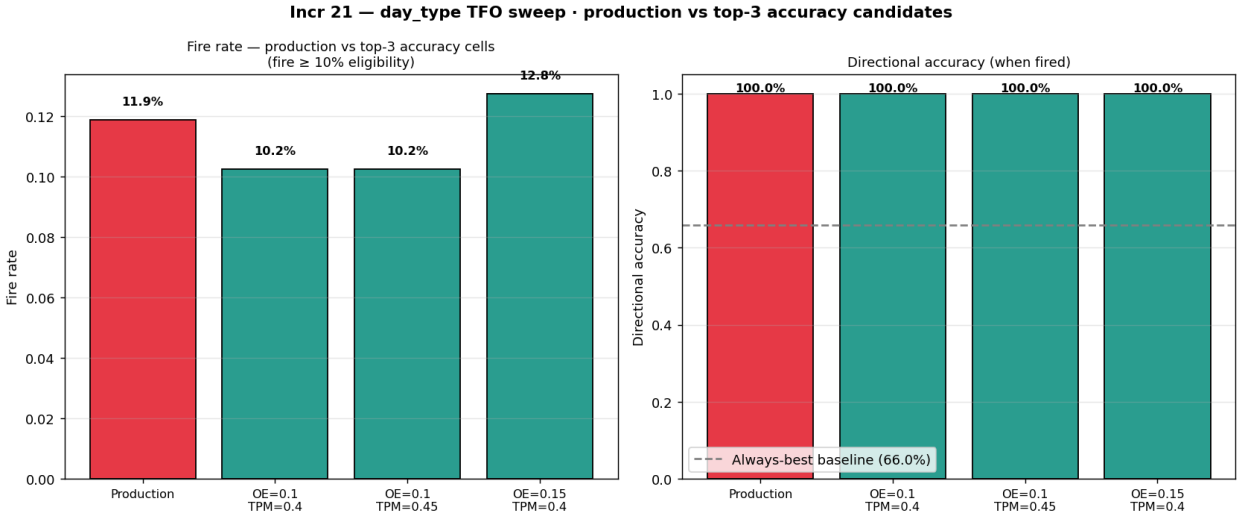
Three heatmaps — fire rate (left), directional accuracy (middle), median |score| (right). Rows = OPEN_EXTREME_THRESHOLD, columns = TREND_PCT_MIN. Red box = production cell. Middle panel is the story — every cell sits at 100% accuracy. Fire-rate panel (left) is the real lever: loosening OE moves coverage monotonically.

Pareto view — fire rate vs accuracy



Every (OE, TPM) cell plotted. Dashed grey = always-best baseline; all 20 cells sit at the top of the chart (100% accuracy). No tradeoff between coverage and accuracy in this sweep — the gate dominates its baseline no matter which cell you pick.

Production vs top-fire-rate candidates (fire ≥ 10%)



Since every cell ties at 100% accuracy, the ranking collapses to fire rate. Red bar = production; green bars = top-3 fire-rate cells. +3-4 pp coverage available at zero accuracy cost by loosening OE.

What this implies

- **TFO-fires sessions carry the cleanest directional signal in the stack.** When TFO fires, the session is mathematically bound to close in its direction. Equal-weighting in the aggregate dilutes this 1:11 — candidate for a weight > 1.0 on fires.
- **The 19.6% day_type fire rate from incr 19 is NOT explained by a too-strict TFO gate alone.** TFO fires at 12%, so the remaining 7-8 pp come from spike_and_channel, trending_tr, and the trading_range → trend_from_open chop-ratio override. Those branches weren't swept here.
- **Coverage is the only lever.** Looser cells strictly dominate stricter ones — no accuracy tradeoff in this sample.

What's next — needs your nod

- **NEW CANDIDATE:** loosen **OPEN_EXTREME_THRESHOLD: 0.15 → 0.25** in aiedge/context/daytype.py::_classify_day_type (currently a magic literal). Gains 2-3 pp TFO fire rate with zero accuracy cost. Small change; lowest-risk of the pending items.
- **STILL PENDING (incr 20):** change ALWAYS_IN_WINDOW: 5 → 10 and mirror STRONG_TREND_WINDOW in signals/components.py. Independently motivated; incr 20 showed +16 pp fire / +9 pp accuracy.
- **STILL PENDING (incr 18):** lower MAJORITY_TREND_BAR_FLOOR to 0.25.
- **STILL PENDING (incr 19):** docstring patch on compute_trend_state flagging the silent-zero htf_alignment trap.
- **NEXT STUDY (read-only):** full day_type sweep including the spike_and_channel and trading_range branches. Requires parameterising the whole classifier.
- **Caveat:** 'directional accuracy' here is same-session open→close labelling fidelity, not forward WR. Pattern Lab WR-driven confirmation is the gold standard for weighting changes.

Findings site: https://github.com/zerosumsystems-ui/iPhone-/tree/main/trend-classification